Data Validation and Behavioral Stability

In response to the reviewer's suggestion, we conducted a follow-up one-month field observation (Monday to Friday, October–November 2025) to further validate key simulation parameters regarding charging station configuration, vehicle composition, and vehicle behaviour.

1. Validation of Value Range in Simulation Scenarios

Results show that the number of active charging points increased to **29 at Building C** and **27 at Building J**, remaining within the **mid-range of the simulated scenarios (up to 50 ports)**. This confirms the representativeness of the simulation setup and validates the scalability assumptions tested for future expansion scenarios.

Weekly Charging Infrastructure Check

Week	Building	Total active charging ports	11kW ports	30kW ports
Week 1 (1/10)	C	29	24	5
Week 2 (6/10)	C	29	24	5
Week 3 (13/10)	C	29	24	5
Week 4 (20/10)	C	29	24	5
Week 5 (27/10)	C	29	24	5

Week	Building	Total active charging ports	11kW ports	30kW ports
Week 1 (1/10)	J	27	23	4
Week 2 (6/10)	J	27	23	4
Week 3 (13/10)	J	27	23	4
Week 4 (20/10)	J	27	23	4
Week 5 (27/10)	J	27	23	4

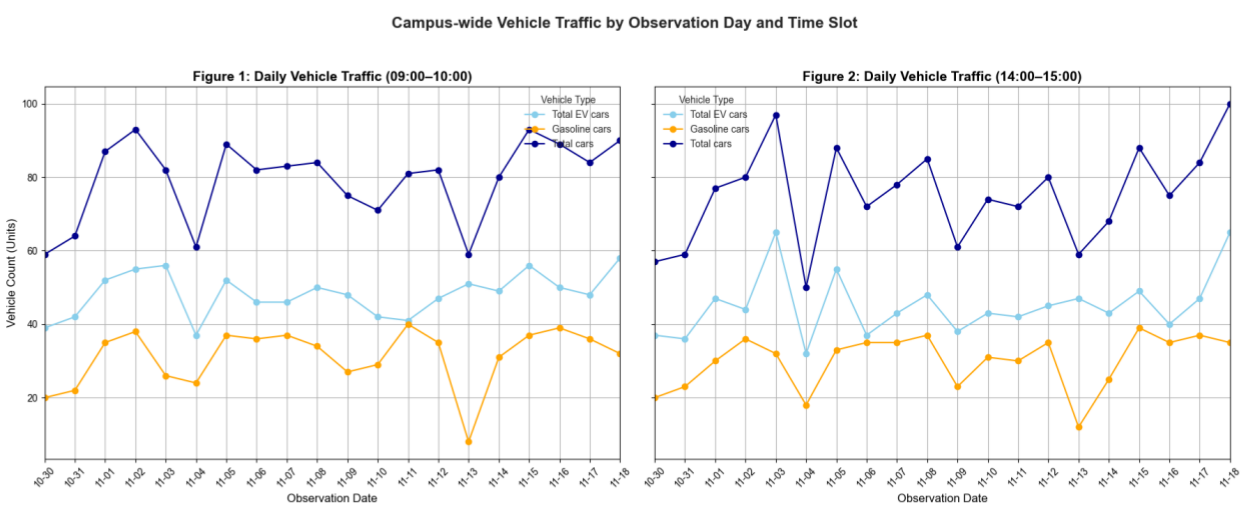
Table 6: Value ranges for various configurations of charging stations and solar panels, including options for C-Parking and J-Parking.

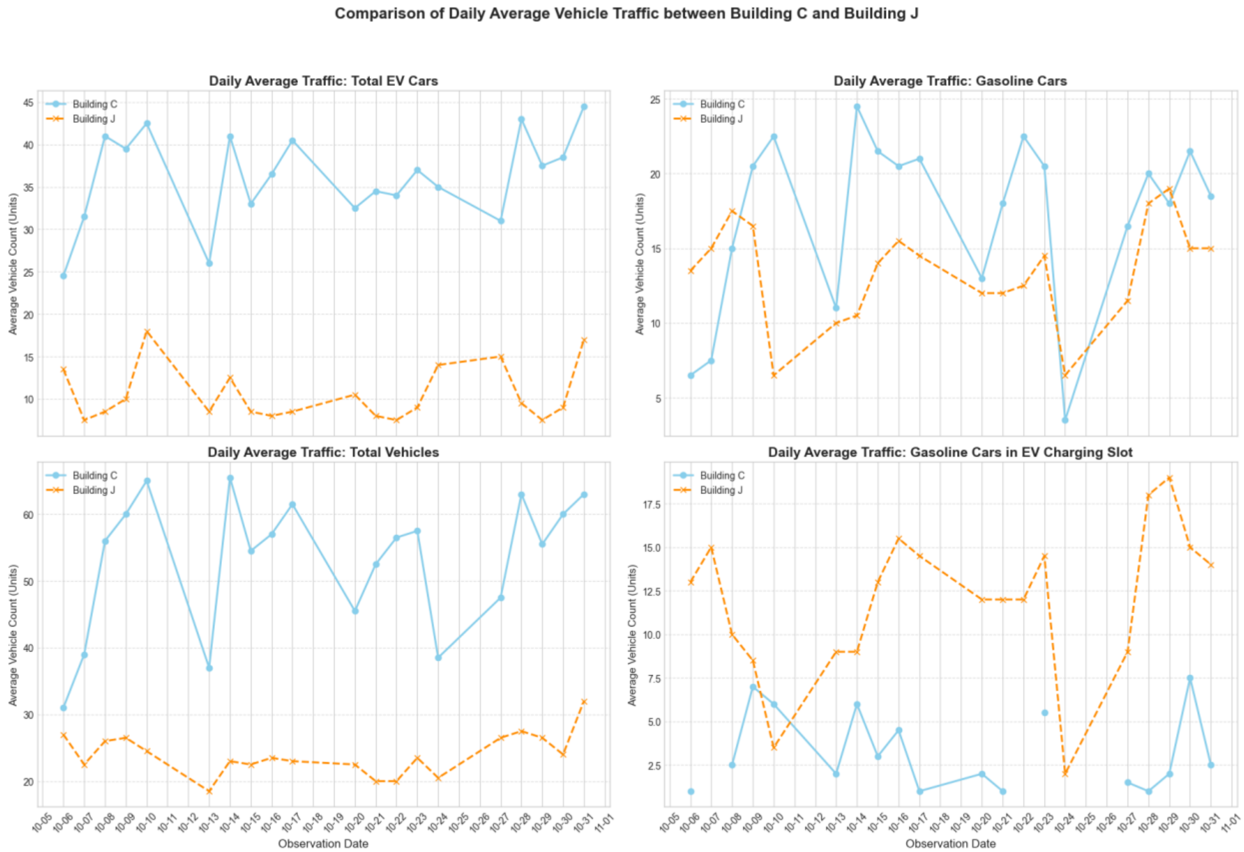
Category	Configuration Options
C-Parking	11kW Charging Ports: 20, 25, 30, 35, 40, 45, 50
	30kW Charging Ports: 2, 4, 6, 8, 10
J-Parking	11kW Charging Ports: 15, 18, 21, 24, 27, 30
	30kW Charging Ports: 2, 4, 6, 8
Solar Panels	200, 300, 400, 500, 600, 700, 800, 900
BESS capacity	80kW
Weather Data	Annual average dataset

The **number of gasoline vehicles remained stable at approximately 30 units**, reinforcing the assumption of a fixed gasoline-vehicle baseline across all simulations and reflecting the ongoing transition toward electric vehicles. Meanwhile, **EV counts fluctuated between 30 and 65**, consistent with the **first two simulated demand scenarios** (50–100 EVs).

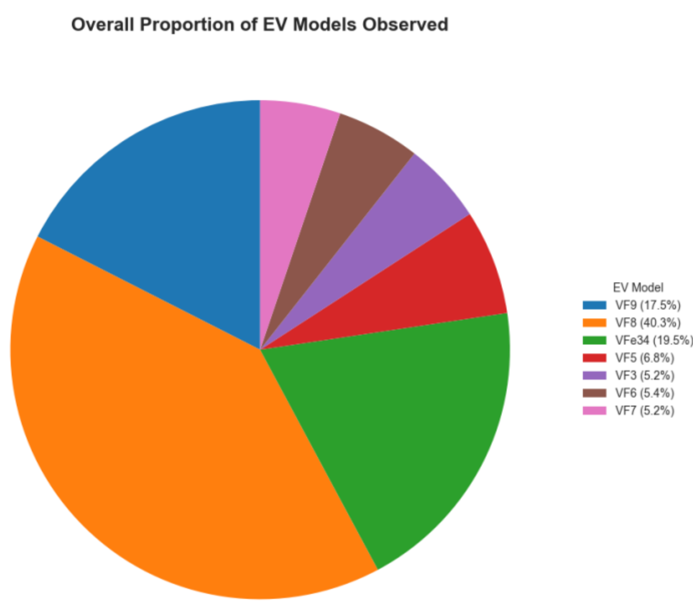
	Total EV cars	Gasoline cars	Total cars	Gasoline cars parked in active CP slot
count	40.00	40.00	40.00	40.00
mean	46.70	30.60	77.30	14.55
std	7.45	7.64	12.29	5.00
min	32.00	8.00	50.00	2.00
25%	42.00	25.75	70.25	13.00
50%	47.00	33.50	80.00	14.00
75%	50.25	36.00	85.50	18.00
max	65.00	40.00	100.00	24.00

In addition, the **spatial distribution of vehicles between Buildings C and J** remained consistence (with Building C as the dominant destination), as in the original setup, and the **observed parking ratios of arrival between morning (9 AM) and afternoon (2 PM)** showed minimal variation. This consistency further validates the temporal and spatial assumptions used for calibration, supporting the representativeness of the simulation inputs and reinforcing behavioral stability across observation periods.





The **share of new EV models** (e.g., VF3, VF6, VF7, VF5) increased slightly, about 20 % of the total, but the **dominant models (VF8, VF9, VFe34)** remained unchanged. New EV configurations can also be easily added in future experiments.



This stability indicates that the simulation structure assumptions remain robust and that the predefined value ranges used for agents and scenario configurations continue to capture current real-world conditions. It also confirms that new EV configurations and updated infrastructure parameters can be readily integrated into the framework, reinforcing its suitability as a decision-support and planning tool for future infrastructure expansion.

2. Validation of Behavioral Stability

2.1. Survey Design and Sampling

The survey was designed to capture realistic mobility and charging behaviors of three primary user groups at VinUniversity, faculty, staff, and students. The questionnaire included three sections:

1. **Demographics and travel schedules:** Arrival and departure times, trip frequency, and weekday regularity.

2. **Parking behavior:** Preferences and compliance between EV and gasoline vehicle users, including parking duration and rule adherence.

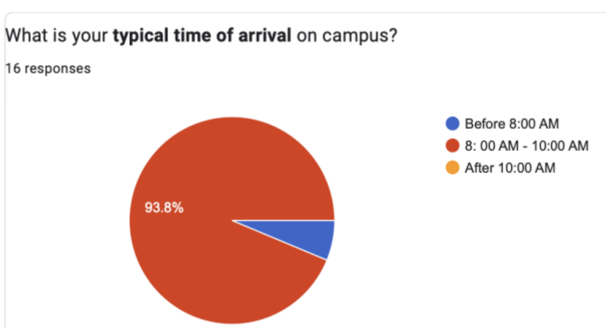
3. **EV charging behavior:** Charging frequency, State of Charge (SoC) triggers, influencing factors (location, speed, schedule), and post-charging actions.

The updated follow-up survey (October–November 2025, $n = 46$, including 34 EV users) reaffirmed behavioral consistency with the 2023–2024 baseline dataset, confirming stability in weekday mobility patterns and charging habits within the campus.

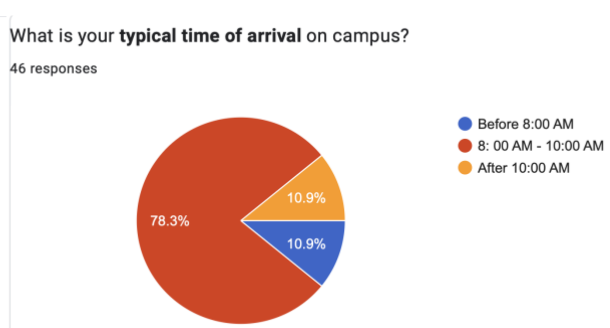
Full details are shown in two reports, "(202510) Electric Vehicles (EV) Charging Behavior Survey" and "(202311) Electric Vehicles (EV) Charging Behavior Survey."

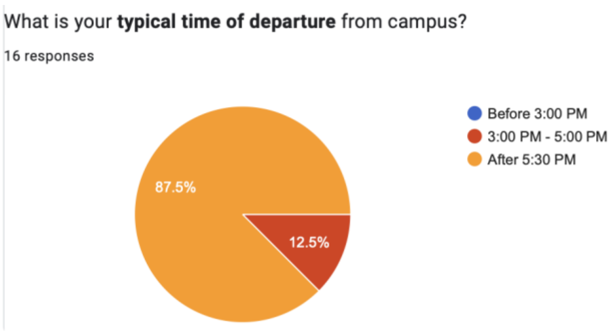
For example:

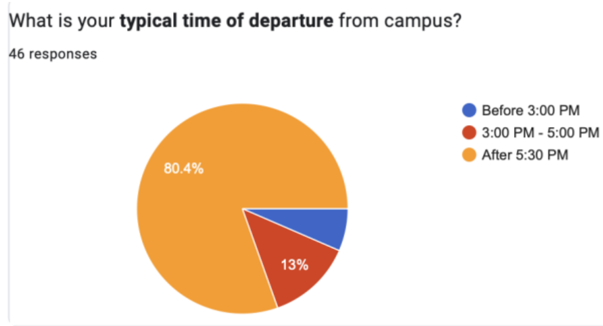
202311 Survey

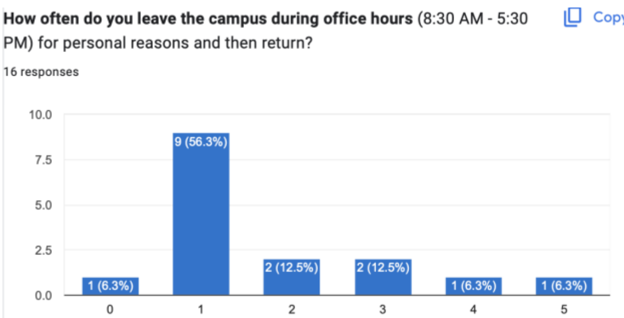


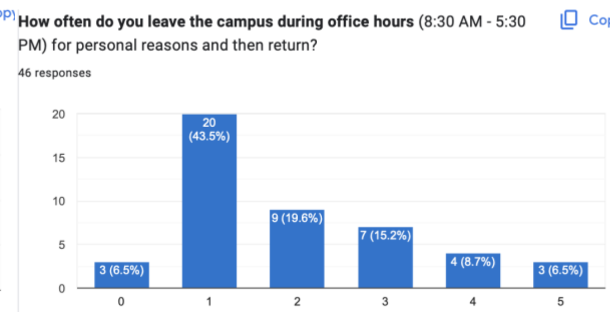
202510 Follow-up Survey











2.2 Integration of Survey Findings into the Model

We reviewed the updated survey results and validated the current model parameters to ensure realistic behavioral diversity and stochasticity. Each behavioral variable was parameterized based on empirical distributions or probabilistic rules, and the results confirmed consistency between survey findings and simulation assumptions.

Behavioral Aspect	Survey Findings (2025 Update)	Integration in Simulation Model
Arrival–Departure Schedule	78% arrive between 8:00–10:00 AM, and 80% depart after 5:30 PM;	Used to generate time-of-day distributions for arrivals and departures, split into morning and afternoon peaks. The current model sets 70% of vehicles starting work between 8–9 AM and leaving after 5 PM.
Random Midday Movement	About 43% leave and return once during office hours; 20% do so twice.	Reconfirmed that the random relocation function triggers movement probability once per day per agent, ensuring dynamic turnover of parking slots.

Charging Trigger (SoC)	56% begin charging when SoC < 35%; 23.5% between 36–50%; 14.7% between 51–70%.	Reconfirmed SoC thresholds that activate charging behavior. The model currently assumes all vehicles charge when SoC < 35% and 80% charge when SoC is between 36–70%.
Charging Preference (Speed)	~60% indicate a low likelihood (≤2/5 on Likert scale) of choosing a distant fast-charging station.	"Priority Fast" flag assigned with 40% probability among all EVs, reflecting a moderate influence of charging speed on decision-making.
Parking Preference (Location)	84% prefer parking close to the destination (≥3/5), and 93% confirm distance to the charging port influences choice (≥3/5).	"Priority Destination" flag assigned with 90% probability, driving parking and charging-slot selection.
Post-charging Relocation	~70% do not move vehicles after a full charge (either unplugging or keeping plugged).	Modeled as a low post-charging relocation rate, reinforcing the policy relevance of parking enforcement scenarios.
Charging Frequency	41% charge once weekly, 26.5% charge 2–3 days per week, and 20.6% 4–6 days per week.	High frequency of charge, so reconfirm the idea for model resets SoC daily to maintain stochasticity and reflect long-term variability in energy-demand cycles.

The updated survey results confirm that the behavioral assumptions and value ranges used in the simulation remain valid and representative of actual user behavior. Minor differences (e.g., in SoC thresholds or charging frequency) fall within the modeled stochastic range, ensuring the framework continues to realistically capture driver heterogeneity and decision-making variability.